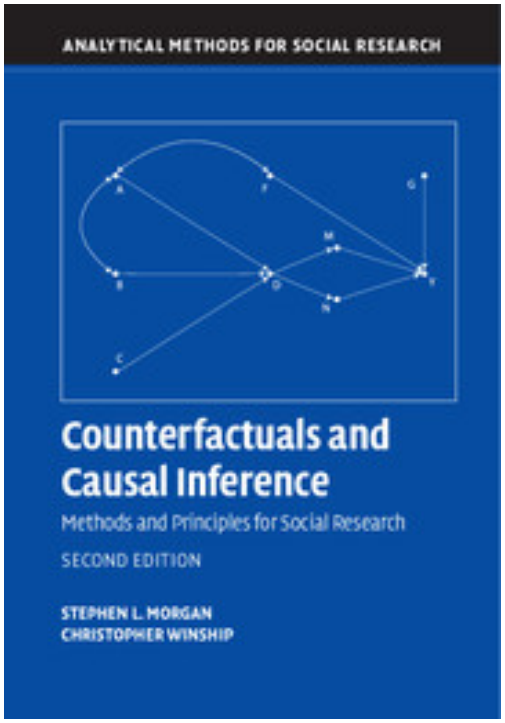
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 12

Outline

1. Distributional Assumptions and Latent Variable Selection-Bias Models
2. Set Identification with Minimal Assumptions
3. Sensitivity Analysis for Provisional Causal Effect Estimates
4. Conclusions

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- Suppose that we are interested in estimating the ATE for the effect of a binary cause D on an interval-scaled outcome Y. Recall the treatment selection model from the econometric tradition:

$$\tilde{D} = Z\phi + U \tag{12.1}$$

where

  - $\tilde{D}$ is a latent continuous variable,
  - Z represents observed variables that determine treatment selection,
  - ϕ is a coefficient (or a vector of coefficients if Z represents more than one variable),
  - U represents both systematic unobserved determinants of treatment selection as well as completely random idiosyncratic determinants of treatment selection.

- Rather than focus on the conditional probability of treatment selection, Heckman instead focused on the conditional expectation function of the latent variable $\tilde{D}$.
- Using the linearity of Equation (12.1), he wrote the conditional expectation function as

$$E[\tilde{D}|Z\phi, D] = Z\phi + E[U|Z\phi, D] \tag{12.2}$$

- If $f(.)$ is the normal density function and $F(.)$ is the corresponding cumulative distribution function, then standard results for truncated normal distributions yield the following expressions for the conditional expectation functions for U:

$$E[U|Z\phi, D = 1] = \frac{f(Z\phi)}{F(Z\phi)} \tag{12.3}$$

$$E[U|Z\phi, D = 0] = \frac{-f(Z\phi)}{[1 - F(Z\phi)]} \tag{12.4}$$

- Equation (12.3) simply gives the formula for Heckman's λ in a standard sample selection problem. In the treatment effects context, this expression is the λ for the treatment group ($D = 1$). The full dummy endogenous variable model requires a second λ for those in the control group ($D = 0$), which is calculated from Equation (12.4).
- These two λ's are then used to form two new regressors,

$$d_i \times \left(\frac{f(z_i\hat{\phi})}{F(z_i\hat{\phi})} \right) \quad \text{and} \quad (1 - d_i) \times \left(\frac{-f(z_i\hat{\phi})}{[1 - F(z_i\hat{\phi})]} \right) \qquad (12.5)$$

- This procedure is identical to Heckman's general lambda method for correcting for selection bias, except that two distinct λ's are utilized, each of which applies either to the treatment group or to the control group.
- Because these supplementary adjustment terms are functions in $z_i\hat{\phi}$ and d_i, the model is often referred to as a control function estimator.
- Estimates from these models can be very sensitive to assumptions about the distribution of U.
- Control function estimators are most effective when the variables in Z include instrumental variables, so that the added regressors in Equation (12.5) have an exogenous source of variation unrelated to the distributional assumption placed on U.

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Manski's approach

- If one cannot (or does not) impose a strong assumption to point-identify and then estimate an average treatment effect, it is natural to wonder about an opposing strategy for analysis: What can be learned about average causal effects while imposing weaker but defendable assumptions?
- In a series of articles and books that have built on less formalized work of the past, Manski has investigated plausible values for treatment effect parameters, such as the ATE, that are consistent with the data when weak assumptions alone are maintained.
- We use the label of "set identification" in contrast to point identification of single values. Manski often uses the alternative phrase "partial identification," and many writers refer to this strategy as an "analysis of bounds."

"No-assumptions" bounds

- Consider a hypothetical example in which we know that both Y^1 and Y^0 are bounded by 0 and 1, and thus that Y is also bounded by 0 and 1.
- The maximum ATE of 1 would occur if $E[Y^1|D=1] = E[Y^1|D=0] = 1$ and $E[Y^0|D=1] = E[Y^0|D=0] = 0$.
- The minimum ATE of -1 would occur instead if $E[Y^1|D=1] = E[Y^1|D=0] = 0$ and $E[Y^0|D=1] = E[Y^0|D=0] = 1$.

From the bound on the outcome, we can conclude

- $E[\delta]$ must be contained in an interval of length 2, which can be stated as a known bound on $E[\delta]$, using the notation

$$-1 \leq E[\delta] \leq 1 \tag{12.6}$$

or that $E[\delta]$ lies in an interval with closed bounds, as in

$$E[\delta] \in [-1, 1]. \tag{12.7}$$

Table 12.1 A Hypothetical Example of the Calculation of Bounds for the ATE

	$E[Y^1 \mid .]$	$E[Y^0 \mid .]$
Naive estimator suggests $E[\delta] = .4$		
Treatment group	$E[Y^1 \mid D = 1] = .7$	$E[Y^0 \mid D = 1] = ?$
Control group	$E[Y^1 \mid D = 0] = ?$	$E[Y^0 \mid D = 0] = .3$
Largest possible $E[\delta] = .7$		
Treatment group	$E[Y^1 \mid D = 1] = .7$	$E[Y^0 \mid D = 1] = 0$
Control group	$E[Y^1 \mid D = 0] = 1$	$E[Y^0 \mid D = 0] = .3$
Smallest possible $E[\delta] = -.3$		
Treatment group	$E[Y^1 \mid D = 1] = .7$	$E[Y^0 \mid D = 1] = 1$
Control group	$E[Y^1 \mid D = 0] = 0$	$E[Y^0 \mid D = 0] = .3$

- If half of the sample is in the treatment group, and if $E[Y^1|D=1] = .7$ and $E[Y^0|D=0] = .3$, then the largest possible treatment effect is .7, whereas the smallest possible treatment effect is $-.3$.
- This result is a straightforward calculation using what-if values for the decomposition of the ATE presented in Equation (2.10), which was used in Chapter 2 and elsewhere to discuss the bias of the naive estimator:

$$\begin{aligned} E[\delta] &= \{\pi E[Y^1|D=1] + (1-\pi)E[Y^1|D=0]\} \quad (12.8) \\ &\quad - \{\pi E[Y^0|D=1] + (1-\pi)E[Y^0|D=0]\}. \end{aligned}$$

- Plugging in the values from the second panel of Table 12.1 into the decomposition in Equation (12.8) yields the largest possible treatment effect as

$$\begin{aligned} E[\delta] &= \{(.5)(.7) + (1 - .5)(1)\} - \{(.5)(0) + (1 - .5)(.3)\} \\ &= .85 - .15 \\ &= .7, \end{aligned}$$

  whereas plugging in the values from the third panel yields the smallest possible treatment effect as

$$\begin{aligned} E[\delta] &= \{(.5)(.7) + (1 - .5)(0)\} - \{(.5)(1) + (1 - .5)(.3)\} \\ &= .35 - .65 \\ &= -.3. \end{aligned}$$

- For this example, $E[\delta] \in [-.3, .7]$, which is an interval of length 1 and is half the length of the maximum interval calculated before estimates of π, $E[Y^1|D = 1]$, and $E[Y^0|D = 0]$ were obtained from the data.

The general case for a 1-0 outcome

- With potential outcome variables bounded by 1 and 0, the no-assumptions bound will always be of length 1.
- The lower bound is derived as

$$\{\pi E[Y^1|D=1]+(1-\pi)(0)\}-\{\pi(1)+(1-\pi)E[Y^0|D=0]\} \tag{12.9}$$

- The upper bound is derived as

$$\{\pi E[Y^1|D=1]+(1-\pi)(1)\}-\{\pi(0)+(1-\pi)E[Y^0|D=0]\} \tag{12.10}$$

- Simplifying and then combining Equations (12.9) and (12.10) yields the no-assumptions bound for potential outcomes bounded by 1 and 0:

$$\begin{aligned} &\pi E[Y^1|D=1] - (1-\pi)E[Y^0|D=0] - \pi \\ &\quad \leq E[\delta] \\ &\quad \leq \pi E[Y^1|D=1] \ - (1-\pi)E[Y^0|D=0] + (1-\pi). \end{aligned} \tag{12.11}$$

The general case for a bounded outcome

- If Y^1 and Y^0 are bounded by any finite values a and b, where $b > a$, such as theoretical minimum and maximum scores on a standardized test, then Equation (12.6) can be written more generally as

$$-b + a \leq E[\delta] \leq b - a, \tag{12.12}$$

which then generates the more general no-assumptions bound for the observed data:

$$\begin{aligned} &\pi E[Y^1|D=1] - (1-\pi)E[Y^0|D=0] + a(1-\pi) - b\pi \\ &\leq E[\delta] \\ &\leq \pi E[Y^1|D=1] - (1-\pi)E[Y^0|D=0] + b(1-\pi) - a\pi. \end{aligned} \tag{12.13}$$

Bounds under additional weak assumptions

For Manski, calculation of the no-assumptions bound is only the starting point of a set-identification analysis. The primary goal is to analyze how additional assumptions can narrow the no-assumptions bound:

> *Empirical researchers should be concerned with both the logic and the credibility of their inferences. Credibility is a subjective matter, yet I take there to be wide agreement on a principle I shall call:*
>
> *The Law of Decreasing Credibility: The credibility of inference decreases with the strength of the assumptions maintained.*
>
> *This principle implies that empirical researchers face a dilemma as they decide what assumptions to maintain: Stronger assumptions yield inferences that may be more powerful but less credible. (Manski 2003:1)*

Monotone treatment response (MTR)

- In many situations, it may be reasonable to assume that the individual-level treatment effect cannot be negative, such that $\delta \geq 0$ for every individual i.
- Under MTR (in the direction where $\delta \geq 0$), the lower bound for the ATE must be 0.
- MTR implies that members of the control group have counterfactual values of y_i^1 that are at least as high as their observed values of y_i.
  - If $y_i = y_i^0$ for members of the control group, then the MTR assumption that $y_i^1 \geq y_i^0$ for all individuals i implies that $y_i^1 \geq y_i$ for members of the control group.
  - For the treatment group, the counterfactual values y_i^0 are no higher than the observed values of y_i.

- Under MTR for the hypothetical example presented in Table 12.1, one can therefore replace the extreme values of 1 and 0 in the no-assumptions lower bound:

$$\{(.5)(.7) + (1 - .5)(0)\} \; - \{(.5)1 + (1 - .5)(.3)\} = -.3$$

with less extreme values of .7 and .3:

$$\{(.5)(.7) + (1 - .5)(.3)\} \; - \{(.5)(.7) + (1 - .5)(.3)\} = 0$$

- Assuming MTR in this direction, the bound for ATE is $[0, .7]$ rather than $[-.3, .7]$.

Monotone treatment selection (MTS)

- It may also be reasonable to assume that those who receive the treatment have higher average outcomes under potential exposure to both the treatment and control.
- In most cases, one would assume jointly that $E[Y^1|D=1] \geq E[Y^1|D=0]$ and $E[Y^0|D=1] \geq E[Y^0|D=0]$, which is traditionally thought of as positive self-selection.

- For the hypothetical example presented in Table 12.1, MTS implies that the naive estimator would be an upper bound for the ATE. In short, MTS in this direction stipulates that members of the treatment group could not have done any worse in the control state than those observed in the control group (and, vice versa).
- Maintaining MTS allows one to replace the extreme values of 1 and 0 in the no-assumptions upper bound,

$$\{(.5)(.7) + (1 - .5)(1)\} - \{(.5)0 + (1 - .5)(.3)\} = .7,$$

with the less extreme values of .7 and .3, yielding for this example,

$$\{(.5)(.7) + (1 - .5)(.7)\} - \{(.5)(.3) + (1 - .5)(.3)\} = .4.$$

- MTS thereby implies that the bound for the ATE is $[-.3, .4]$ rather than $[-.3, .7]$.

MTR and MTS together

- The power of Manski's approach comes from the ability to apply multiple assumptions at the same time in order to narrow the no-assumptions bound to a bound that is considerably more informative.
- For the hypothetical example presented in Table 12.1, one can narrow the no-assumptions bound from $[-.3, .7]$ to $[0, .4]$ by invoking the MTR and MTS assumptions together.
- The resulting bound still includes 0, but in some applications such narrowing may well be helpful in ruling out some unreasonable and extreme causal claims.

Monotone instrumental variables (MIV)

- Manski and Pepper (2000) first define the traditional IV assumption in terms of mean independence, after conditioning on a covariate X for generality.
- In our notation, the standard IV assumption of no direct effect of Z on Y (by either Y^1 or Y^0) is written out as

$$E[Y^1|X, Z = z'] = E[Y^1|X, Z = z''], \tag{12.14}$$

$$E[Y^0|X, Z = z'] = E[Y^0|X, Z = z''], \tag{12.15}$$

for any two values z' and z'' of Z (and separately for strata defined by X).

- Manski and Pepper (2000) then consider a weaker assumption for an IV analysis, known as a monotone IV (MIV) assumption.
- For all values of the instrument Z, in which $z'' \geq z'$, the variable Z is an MIV if

$$E[Y^1|X, Z = z''] \geq E[Y^1|X, Z = z'] \tag{12.16}$$

$$E[Y^0|X, Z = z''] \geq E[Y^0|X, Z = z'] \tag{12.17}$$

- In Equations (12.16) and (12.17), the expected values of both potential outcomes are weakly increasing in Z.

- To what extent does an MIV narrow the bound for the ATE (or, in the words of Manski (2003), shrink the identification region for it)? The short answer: No more than a traditional IV (and usually considerable less), but still enough to have some identifying power.
- Manski and Pepper estimate bounds on the effect of education on the logged wages of respondents for the National Longitudinal Survey of Youth.
  - With MTR and MTS, the bound for the effect of a twelfth year of schooling was $[0, .199]$, that the bound for the effect of a fifteenth year of schooling was $[0, .255]$, and that the bound for the effect of a sixteenth year of schooling was $[0, .256]$.
  - When they then used the Armed Forces Qualification Test as an MIV while still maintaining the MTR and MTS assumptions, they obtained narrower bounds, respectively, of $[0, .126]$, $[0, .162]$, and $[0, .167]$.

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Recall a familiar directed graph

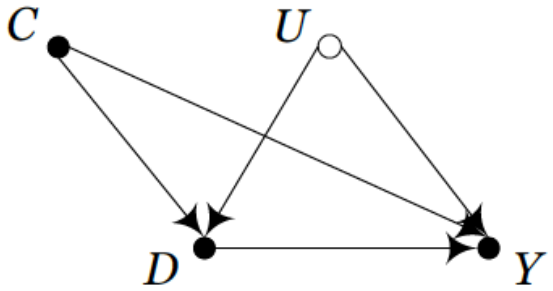

Figure 12.1 A graph in which the causal effect of D on Y is confounded by an observed variable C and an unobserved variable U.

Sensitivity analysis

- Closely related to an analysis of bounds, but also very different.
- Rosenbaum (2002; see also Rosenbaum 2010) devotes a large portion of his excellent book on observational data analysis to strategies for performing sensitivity analysis, especially for matching designs.
- Consider for reference the simple directed graph in Figure 12.1, where an analyst seeks to estimate the effect of D on Y. In this case, the researcher observes only one of two known confounders, C but not U.
- For Rosenbaum, adjusting for C would remove "overt bias" but would leave the "hidden bias" produced by $D \leftarrow U \rightarrow Y$ untouched.

- For a sensitivity analysis, the typical steps would be
  1. Offer a provisional point estimate of the ATE for the effect of D on Y by conditioning on C and assuming that ignorability holds.
  2. Choose multiple pairs of values for the effects of U on D and of U on Y.
  3. Using either analytic methods or simulation methods, present results that show how much the provisional point estimate of the ATE would be expected to change assuming each pair of values.
  4. Identify and report the pairs of values that would push the provisional point estimate of the ATE below the critical level that would prevent one from rejecting the null hypothesis of no effect.
  5. Use external information on what is known about U and its relationships to D and Y to assess whether or not the pairs of values that would prevent one from rejecting the null hypothesis of no effect are reasonably likely.

- If, after after executing these steps, the researcher can argue that all plausible violations of ignorability (i.e., the pairs of values for the effects of U on D and of U on Y) would not change the qualitative conclusions, then one can report the levels of sensitivity from step 3 and confidently stick to the main substantive conclusion that the ATE is sufficiently unlikely to be equal to 0.
- If, however, plausible violations of ignorability would change the main substantive conclusion, because 0 cannot be regarded as implausible, then one should step back from strong conclusions and consider other methods of analysis.

- Note how sensitivity analysis differs from the set-identification approach presented above.
  - From this alternative orientation, the researcher would first calculate no-assumptions bounds for the conditional average effect of D on Y within strata of C.
  - Then, the researcher would introduce additional credible assumptions about the individual-level and/or average effects of U on D and U on Y in an attempt to narrow the no-assumptions bounds to intervals that all researchers can accept.

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1. Selection-bias and control function estimators: Because distributional assumptions are hard to justify, these are rarely used. They are now used to justify IV estimation, and to promote the local IV literature that Heckman and his colleagues have developed recently.
2. Set-identification: Rarely used because they typically cannot eliminate 0 as a permissible value for estimated causal effects and, even more so, because social scientists continue to very much prefer single-value estimates.
3. Sensitivity analysis: More frequently used, but sometimes the efforts are facile and reinforce an unhealthy obsession with the rejection of null hypotheses.